Q&A

Things You Wanted to Know!

Q1. What is the difference between MATLAB and Vivado? Do we need both MATLAB and Vivado Design Suite for programming a ZCU111 Evaluation Kit?

Pournima

A1. There are many differences between MATLAB and Vivado. While MATLAB is a numerical computing and programming language offered by MathWorks, Vivado is a design automation tool for embedded systems offered by Xilinx. Vivado is targeted at larger FPGAs and is slowly replacing Xilinx ISE as their mainline tool chain. Vivado tool has a wide range of applications for designers especially in embedded systems and RF domains.

I think you are talking about Zynq UltraScale+ RFSoC ZCU111 Evaluation Kit which provides a rapid, comprehensive RF analogue-to-digital signal chain prototyping platform. This Evaluation Kit supports Vivado Design Suite as well as MATLAB. It features system-level design using signal capture and analysis with MATLAB and Simulink. You can work on certain applications with Evaluation Kit using MATLAB. But to explore the full potential of this kit, you may need both MATLAB and Vivado Design Suite.

For more details on MATLAB and Vivado Design Suite, you may visit the following websites:

https://www.mathworks.com/products/connections/product_detail/avnet-rfsoc-dev-kit.html and

https://forums.xilinx.com/t5/Xilinx-Evaluation-Boards/Problem-with-Hardware-Co-Simulation-for-ZCU111-board/td-p/988291

Q2. What is e-paper display? What are its advantages and disadvantages?

A. Samiuddhin

A2. E-paper or electronic paper is a special type of display that does not need electricity to sustain the displayed text and image. Just like real paper, the displayed image resembles ink on paper (hence the name). Unlike traditional LCD or OLED display, it does not emit light but reflects light. This characteristic makes e-paper display very comfortable to read even under direct sunlight (refer Fig. 1). Unlike LCD or TFT display, e-paper also does not require a backlight.

The displays are made of millions of minuscule capsules filled with a clear fluid containing microscopic particles of different colours and electrical charges (refer Fig. 2). Electrodes located above and below the capsules move up and down when a positive or negative electric field is applied, making the surface to reflect a certain colour.

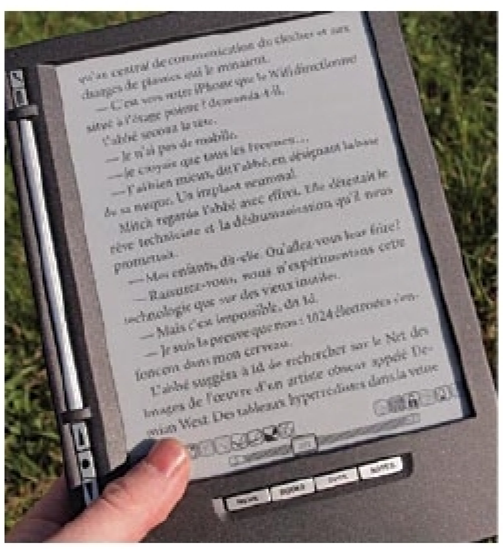

Fig. 1: e-paper based e-book reader visible in the sunlight (Credit: wikipedia.org)

The displays can hold static text and images for months without electricity. That is, it can retain text and images even when power is off.

However, it has many disadvantages as well. That include its high cost as compared to other displays, slow to update text and images, less memory, among others.

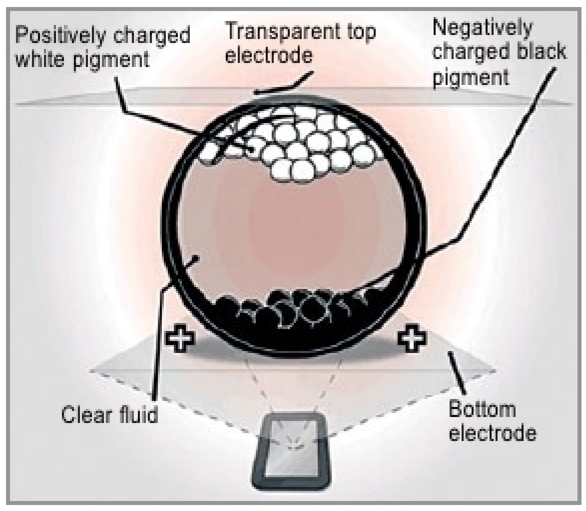

Fig. 2: Scheme of e-paper display (Credit: smartcity-displays.com)

Advantages.

1. Consumes less power
2. Display stays for a long time without power
3. Ideal for low-power projects including Arduino, NodeMCU and other embedded prototyping boards
4. It is available in multiple sizes
5. It is a high-definition and paper-like appearance display

Disadvantages.

1. Display is slow to update
2. Not suitable for animation or fast-changing displays
3. Costlier as compared to LCD and other similar size displays.
4. Currently, it has limited software library and supports very few fonts
5. It has less memory

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